

Frontend Engineering Fundamentals: A 6-Week Intensive Course

WEEK 4: Advanced Git, GitHub & Introduction to React

Learning Objectives

By the end of this week, students will:

- Understand version control with Git and GitHub
- Learn React fundamentals (components, props, state)
- Build functional components with hooks
- Understand component lifecycle
- Manage application state effectively

Learning Materials

Git Advanced & GitHub

Videos:

- "Git and GitHub for Beginners" by freeCodeCamp (1 hour)
<https://www.youtube.com/watch?v=RG0j5yH7evk>
- "Git Branches Tutorial" by Traversy Media (30 mins)
<https://www.youtube.com/watch?v=e2IbNHi4uCI>
- "Git and GitHub Tutorial" by The Net Ninja
https://www.youtube.com/watch?v=3RjQznt-8kE&list=PL4cUxeGkcC9goXbgTDQ0n_4TBzOO0ocPR

Reading:

- Git Documentation - Getting Started
<https://git-scm.com/book/en/v2/Getting-Started-About-Version-Control>

Practice:

- Create repository, make commits, create branches, merge
- Push all previous projects to GitHub with proper README files

React Fundamentals

Videos:

- "React Tutorial for Beginners" by Programming with Mosh (1 hour)
<https://www.youtube.com/watch?v=SqcY0GlETPk>
- "React Hooks Course" by Web Dev Simplified (1 hour)
<https://www.youtube.com/watch?v=O6P86uwfdR0>
- "useState and useEffect Explained" by Fireship (10 mins)
<https://www.youtube.com/watch?v=O6P86uwfdR0>
- "React in 100 Seconds" by Fireship
<https://www.youtube.com/watch?v=Tn6-PIqc4UM>

Reading:

- React Official Docs - Quick Start
<https://react.dev/learn>
- React Official Docs - Thinking in React
<https://react.dev/learn/thinking-in-react>

Practice: Convert Week 2 Task Manager to React

React State Management

Videos:

- "React State Management" by Traversy Media (45 mins)
<https://www.youtube.com/watch?v=35lXWvCuM8o>
- "Props vs State in React" by Web Dev Simplified (15 mins)
<https://www.youtube.com/watch?v=IYvD9oBCuJI>
- "useEffect Hook Explained" by Web Dev Simplified
<https://www.youtube.com/watch?v=0ZJgIjluY7U>

Reading:

- React Docs - Managing State
<https://react.dev/learn/managing-state>
- React Docs - Sharing State Between Components
<https://react.dev/learn/sharing-state-between-components>

Practice: Build a counter app, then a shopping cart component

Week 4 Challenge Project

Build a Weather Dashboard App in React

Requirements:

- Search for any city's weather ([OpenWeatherMap API](#))
- Display current weather with icons
- Show 5-day forecast
- Recent searches saved in localStorage
- Toggle between Celsius and Fahrenheit
- Loading states and error handling
- Fully responsive design

Deliverable: Live React app (Netlify/Vercel) + GitHub repository

Success Criteria:

- Proper React component architecture
- Effective use of hooks (useState, useEffect)
- Clean, reusable components
- Props and state managed correctly
- Professional Git commit history